AI Ethics Assignment – Part 2: Case Study Analysis

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Case 1: Biased Hiring Tool (Amazon Scenario)

Scenario:
Amazon's AI recruiting tool penalized female candidates because it learned from historical resumes that were predominantly male.

1. Source of Bias

- **Training Data Bias:** The dataset reflected historical male dominance in hiring; the AI learned patterns favoring men.
- **Feature Bias:** Gender-associated features such as names, pronouns, or women's colleges influenced the algorithm negatively.
- **Model Design Bias:** No fairness constraints or human oversight were implemented, allowing biased patterns to persist.

2. Fixes to Make the Tool Fairer

1. **Balance the Training Data:** Include equal representation of genders and diverse backgrounds.
2. **Remove Gender-Correlated Features:** Exclude features that are correlated with gender (names, gendered terms, schools).
3. **Use Fairness-Aware Algorithms:** Apply in-processing techniques like adversarial debiasing or reweighting to reduce bias during training.
4. **Introduce Human Oversight:** Ensure AI recommendations are reviewed by humans before decisions are finalized.

3. Metrics to Evaluate Fairness Post-Correction

Metric	Purpose
Disparate Impact Ratio	Checks whether outcomes are equitable across groups (target ≈ 1).
Equal Opportunity Difference	Measures difference in true positive rates between groups.
Statistical Parity Difference	Compares positive outcome rates between groups.

Case 2: Facial Recognition in Policing

Scenario:
Facial recognition systems misidentify minorities at higher rates, raising ethical and legal concerns.

1. Ethical Risks

- **Wrongful Arrests:** Higher false positives disproportionately affect minority populations.
- **Racial Profiling:** Reinforces systemic biases in policing practices.
- **Privacy Violations:** Individuals are surveilled without consent or knowledge.
- **Accountability Gaps:** Difficult to determine responsibility when AI errors occur.

2. Recommended Policies for Responsible Deployment

1. **Mandatory Bias Audits:** Regularly evaluate error rates across demographic groups.
2. **Human-in-the-Loop Verification:** Require human confirmation before law enforcement actions based on AI predictions.
3. **Deployment Restrictions:** Limit use to high-confidence cases; avoid unsupervised mass surveillance.
4. **Transparency & Reporting:** Publish demographic error rates and system performance metrics publicly.
5. **Consent & Opt-Out:** Allow individuals to know when facial recognition is used and provide opt-out options where feasible.

Reflection:
Both cases demonstrate that AI systems can perpetuate societal biases if unchecked. Proactive fairness auditing, human oversight, and policy controls are essential to mitigate harm while maintaining the benefits of AI technologies.